

Titanic Dataset - Summary of Findings

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Dataset Overview:

- Total entries: 891 passengers
- Key features: Survived, Pclass, Sex, Age, SibSp, Parch, Fare, Embarked, Cabin
- Missing data:
 - Age: 177 missing values
 - Cabin: heavily missing (~77%)
 - Embarked: 2 missing values

1. Survival by Gender:

Women had a much higher survival rate than men. This reflects the "women and children first" rescue policy.

2. Survival by Passenger Class:

1st class passengers had the highest survival rate, and 3rd class the lowest. This suggests that social status and access influenced survival.

3. Survival by Age:

Children, especially under age 10, had a greater chance of survival. Elderly passengers had lower survival rates.

4. Fare vs Survival:

Passengers who paid higher fares tended to survive more often, indicating that class and fare were influential factors.

5. Embarkation Port:

Passengers who boarded at Cherbourg (C) had higher survival rates. Those who boarded at Southampton (S) had lower survival chances.

6. Correlation Insights:

- Survived is negatively correlated with Pclass and positively with Fare.
- Family size (SibSp, Parch) had minor effects, especially for very large families.

Key Takeaways:

Sex, Passenger Class, and Age were the strongest predictors of survival. The Titanic tragedy highlighted the role of social hierarchy in crisis situations.